**<u>Aim 1</u>:** To generate basic signals like unit impulse, unit step, unit ramp signal and exponential signals.

Q1. WAP to generate the following signals using MATLAB:

a) Unit step sequence b) Ramp sequence c) Exponential sequence

d) Generate and plot each of the following sequences over the indicated interval.

**a.** $x(n) = 2\delta(n+2) - \delta(n-4), \quad -5 \le n \le 5.$
**b.** $x(n) = n[u(n) - u(n-10)] + 10e^{-0.3(n-10)}[u(n-10) - u(n-20)], 0 \le n \le 20.$
**c.** $x(n) = \cos(0.04\pi n) + 0.2w(n), \ 0 \le n \le 50$, where $w(n)$ is a Gaussian random sequence with zero mean and unit variance.
**d.** $\tilde{x}(n) = \{..., 5, 4, 3, 2, 1, 5, 4, 3, 2, 1, 5, 4, 3, 2, 1, ...\}; \ -10 \le n \le 9.$

e) Generate the complex-valued signal.

$$x(n) = e^{(-0.1+j0.3)n}, \quad -10 \le n \le 10$$

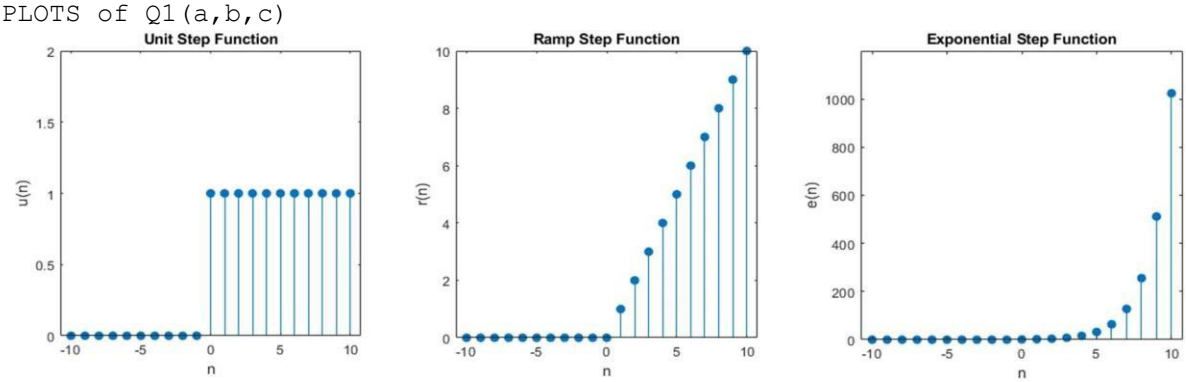
f) Let $x(n) = u(n) - u(n-10)$. Decompose $x(n)$ into even and odd components.

**MATLAB code & plots:**

```
%Part a)
Clear;clc;
n = -10:10;
 u = zeros(1,length(n));

 for k = 1:length(n)
     if n(k) >= 0
         u(k) = 1;
     else
         u(k) = 0;
     end
 end
stem(n,u,'filled'); grid on;
title('Unit Step Sequence'); xlabel('n'); ylabel('u(n)');

%Part b)
Clear;Clc;
n = -10:10;
 r = zeros(1,length(n));

 for k = 1:length(n)
     if n(k) >= 0
         r(k) = n(k);
     else
         r(k) = 0;
     end
 end
stem(n,r,'filled'); grid on;
title('Ramp Sequence'); xlabel('n'); ylabel('r(n)');
```

```
%part c)
Clear;clc;
n = -10:10;
 a = 0.8;
 x = zeros(1,length(n));

 for k = 1:length(n)
     if n(k) >= 0
         x(k) = a^n(k);
     else
         x(k) = 0;
     end
 end

 stem(n,x,'filled'); grid on;
title('Exponential Sequence'); xlabel('n'); ylabel('x(n)');
```

PLOTS of Q1(a,b,c)



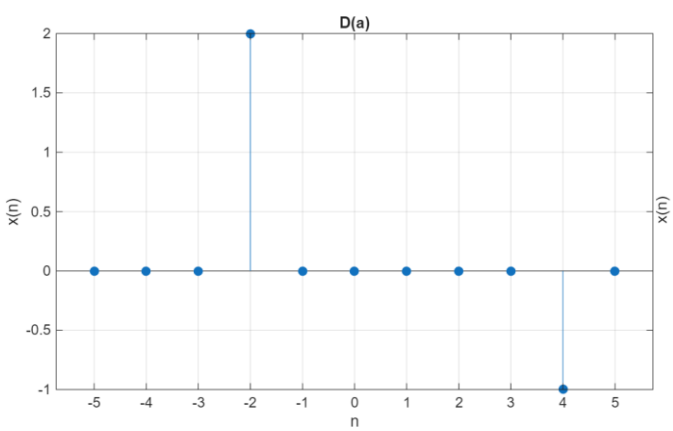

**MATLAB code & plots:**

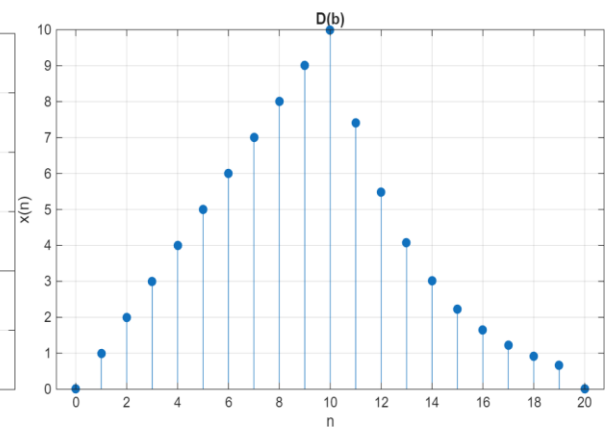
```
%Part d(a)
Clear;clc;
n = -5:5;
 x = zeros(1,length(n));

 for k = 1:length(n)
     if n(k) == -2
         x(k) = 2;
     elseif n(k) == 4
         x(k) = -1;
     else
         x(k) = 0;
     end
 end
stem(n,x,'filled');
xlabel('n'); ylabel('x(n)');
title('D(a)');grid on;
```

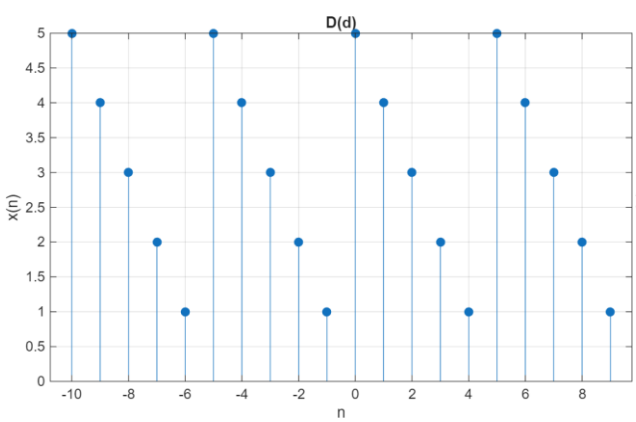
```
%Part d(b)
Clear;clc;
n = 0:20;
 x = zeros(1,length(n));

 for k = 1:length(n)
     if n(k)>=0 && n(k)<10
         part1 = n(k);
     else
         part1 = 0;
     end
     if n(k)>=10 && n(k)<20
         part2 = 10*exp(-0.3*(n(k)-10));
     else
         part2 = 0;
     end
    x(k) = part1 + part2;
 end

 stem(n,x,'filled');
 xlabel('n'); ylabel('x(n)');
 title('D(b)');
 grid on;

%Part d(c)
Clear;clc;
n = 0:50;
 x = zeros(1,length(n));
 w = zeros(1,length(n));

 for k = 1:length(n)
     w(k) = randn;
     x(k) = cos(0.04*pi*n(k)) + 0.2*w(k);
 end
stem(n,x,'filled');
 xlabel('n'); ylabel('x(n)');
 title('D(c));
 grid on;
```

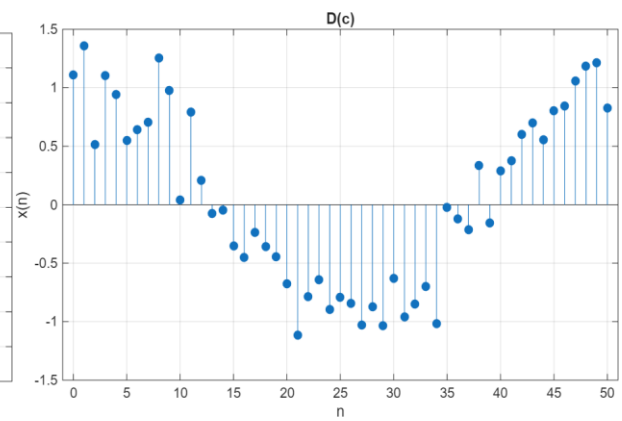
```
%Part  d(d)
Clear;clc;
n = -10:9;
 x = zeros(1,length(n));
 pattern = [5 4 3 2 1];

 for k = 1:length(n)
     index = mod(n(k),5);
     if index < 0
         index = index + 5;
     end
     x(k) = pattern(index + 1);
 end
  stem(n,x,'filled');
 xlabel('n'); ylabel('x(n)');
 title('D(d)');
grid on;
```

PLOTS:-



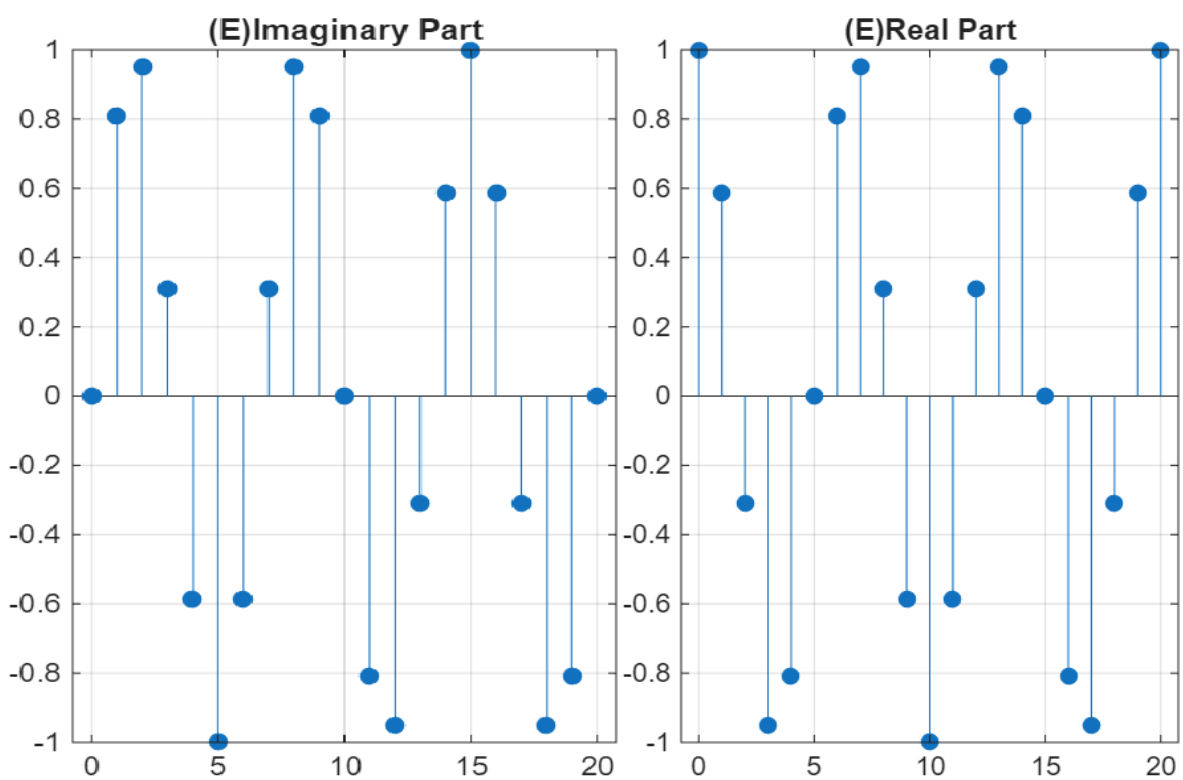



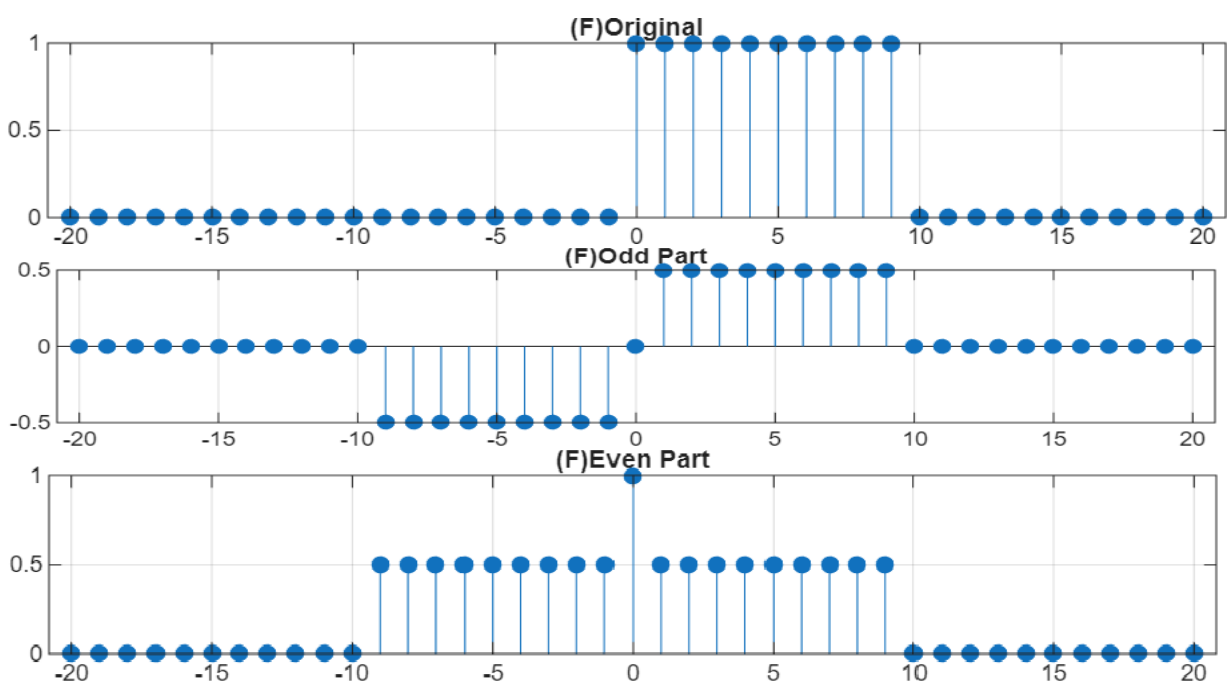
```
%Part e)
Clear;clc;
n = 0:20;
 x = zeros(1,length(n));

for k = 1:length(n)
     x(k) = exp(1j*0.3*pi*n(k));
 end

subplot(1,2,1)
 stem(n,real(x),'filled'); grid on;
 title('(E)Real Part');

subplot(1,2,2)
 stem(n,imag(x),'filled'); grid on;
 title('(E)Imaginary Part');
```

PLOTS:-



```
%Part f)
 n = -20:20;
 x = zeros(1,length(n));
for k = 1:length(n)
     if n(k)>=0 && n(k)<10
         x(k)=1;
     else
         x(k)=0;
     end
 end
x_rev = fliplr(x);
xe = zeros(size(x));
xo = zeros(size(x));
for k = 1:length(x)
     xe(k) = (x(k) + x_rev(k))/2;
     xo(k) = (x(k) - x_rev(k))/2;
 end
subplot(3,1,1)
 stem(n,x,'filled'); title('(F)Original'); grid on;
 subplot(3,1,2)
 stem(n,xe,'filled'); title('(F)Even Part'); grid on;
  subplot(3,1,3)
stem(n,xo,'filled'); title('(F)Odd Part'); grid on;
```

PLOTS:-



Q2. Given the following difference equation

$$y(n) - y(n-1) + 0.9y(n-2) = x(n);\ \forall n$$

**a.** Calculate and plot the impulse response $h(n)$ at $n = -20,.....,100$.

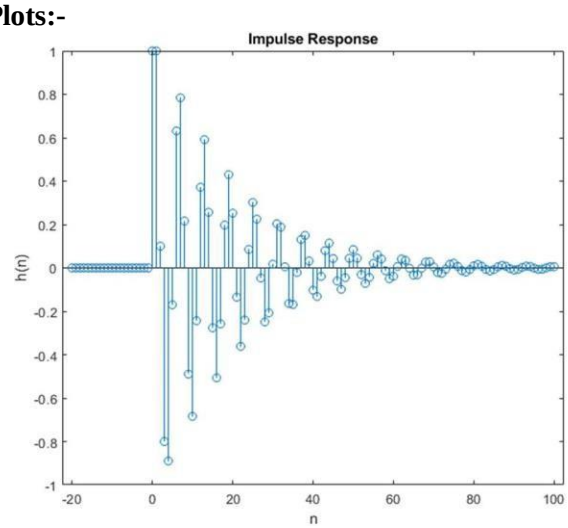
**b.** Calculate and plot the unit step response $s(n)$ at $n = -20,.....,100$.

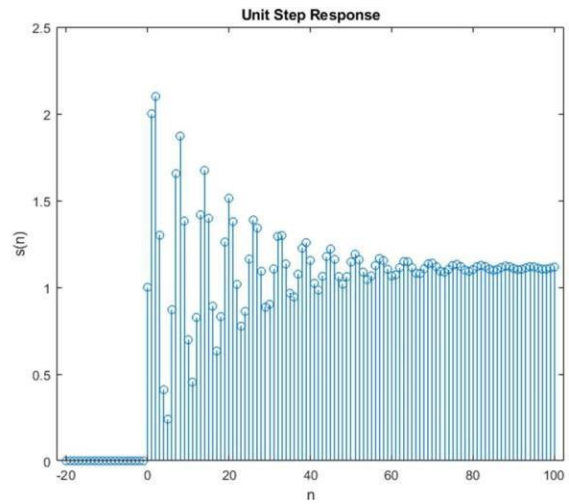
**c.** Is the system specified by $h(n)$ stable?

**MATLAB code & plots:**

```
%Part a)
Clear;clc;
n = -20:100;
x = zeros(1,length(n));
y = zeros(1,length(n));

for k = 1:length(n)
    if n(k)==0
        x(k)=1;
    end
end

for k = 3:length(n)
    y(k) = x(k) + y(k-1) - 0.9*y(k-2);
end

stem(n,y,'filled'); grid on;
title('Impulse Response h(n)');


%Part b)
Clear;clc;
x = zeros(1,length(n));
y = zeros(1,length(n));

for k = 1:length(n)
    if n(k)>=0
        x(k)=1;
    end
end

for k = 3:length(n)
    y(k) = x(k) + y(k-1) - 0.9*y(k-2);
end

stem(n,y,'filled'); grid on;
title('Unit Step Response s(n)');
```

**Plots:-**




c. As the impulse response decays with time and the sum of $h(n)$ is finite. So, the system is stable.

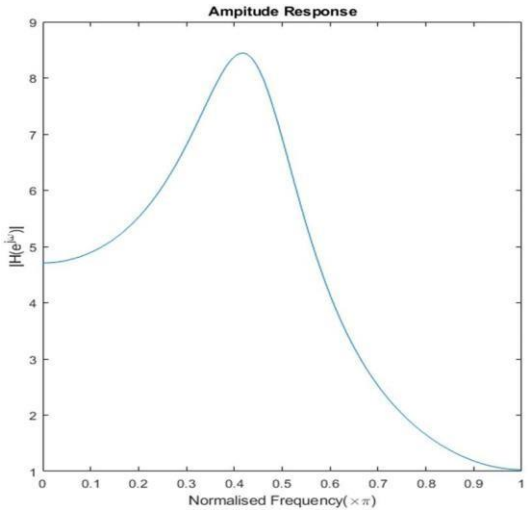
**Aim 2:** To find the frequency response of a given system given in (Transfer Function/ Differential equation form).

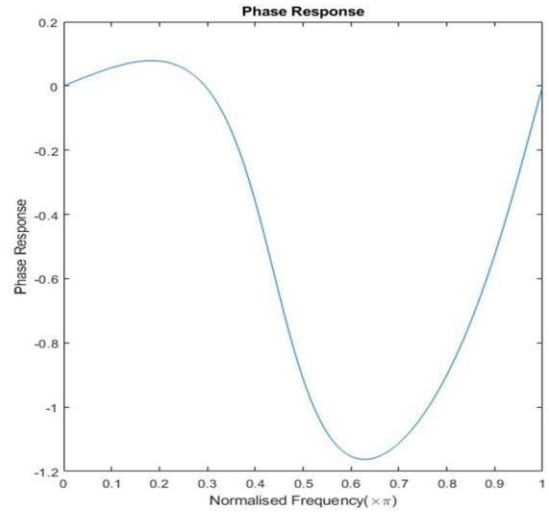
Q3. Find and plot the frequency response (amplitude and phase) of a given system given in (Transfer Function/ Differential equation form).

The difference equation is given as y[n] - 0.25y[n - 1] + 0.45y[n - 2] = 1.55x[n] +1.95x[n - 1] + 2.15x[n]. You can use the inbuilt functions of the MATLAB.

**MATLAB code & plots:**

```
clear; clc;

n = -50:50;

b = [3.7 1.95] ;
a = [1 -0.25 0.45];
[H, w] = freqz(b, a, 512);

subplot(1, 2, 1);
plot(w/pi, abs(H));
xlabel('Normalised Frequency(\times\pi)');
ylabel('|H(e^{j\omega})|');
title('Ampitude Response');

subplot(1, 2, 2);
plot(w/pi, angle(H));
ylabel('Phase Response');
xlabel('Normalised Frequency(\times\pi)');
title('Phase Response');
```



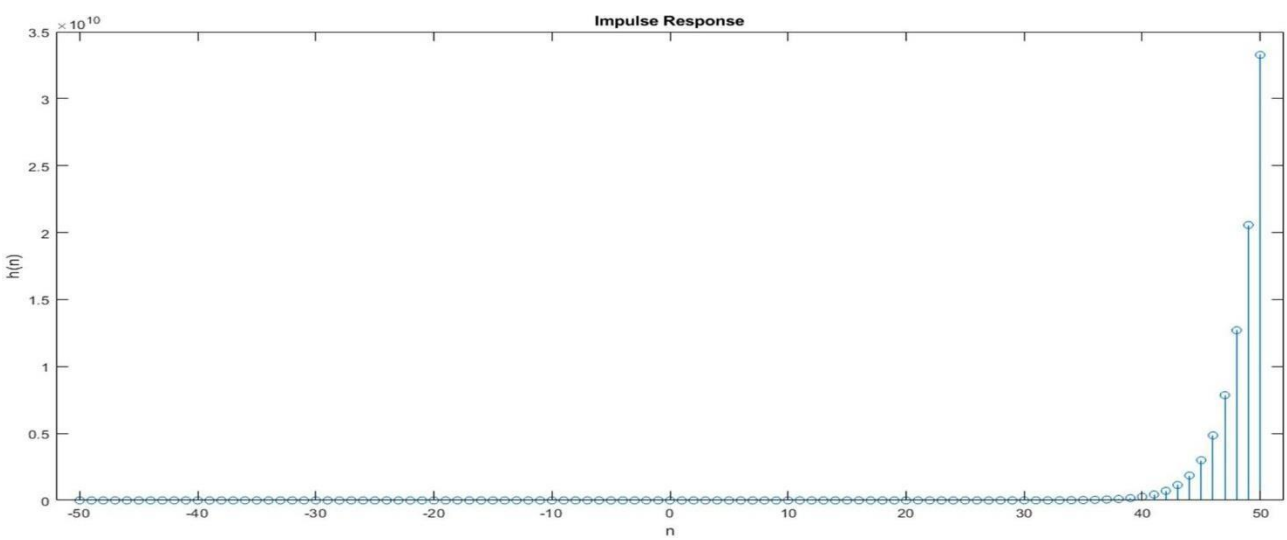

**Aim 3:** To write a MATLAB program to evaluate the impulse response of the system

Q4. Write a MATLAB program to evaluate the impulse response of the system given as:

$$y(n) = x(n)+0.5x(n - 1) + 0.85x(n - 2) + y(n - 1) + y(n - 2).$$

**MATLAB code & plot:**

```
clc; clear;

n = -50:50;
x = double(n == 0);
b = [1 0.5 0.85] ;
a = [1 -1 -1];
h = filter(b, a, x);

stem(n, h);
title('Impulse Response');
xlabel('n');
ylabel('h(n)');
```



**Aim 4:** To write a "C" program to generate various wave forms (Sine wave, Square wave) graphically with using Code Composer Studio (CCS-7.4) IDE and the DSP TMS 320C6713 DSK Kit.

**REQUIREMENT:** 1. Code Composer Studio (Software installed in Desktop PC)

2. DSP TMS 320C6748 DSP Kit

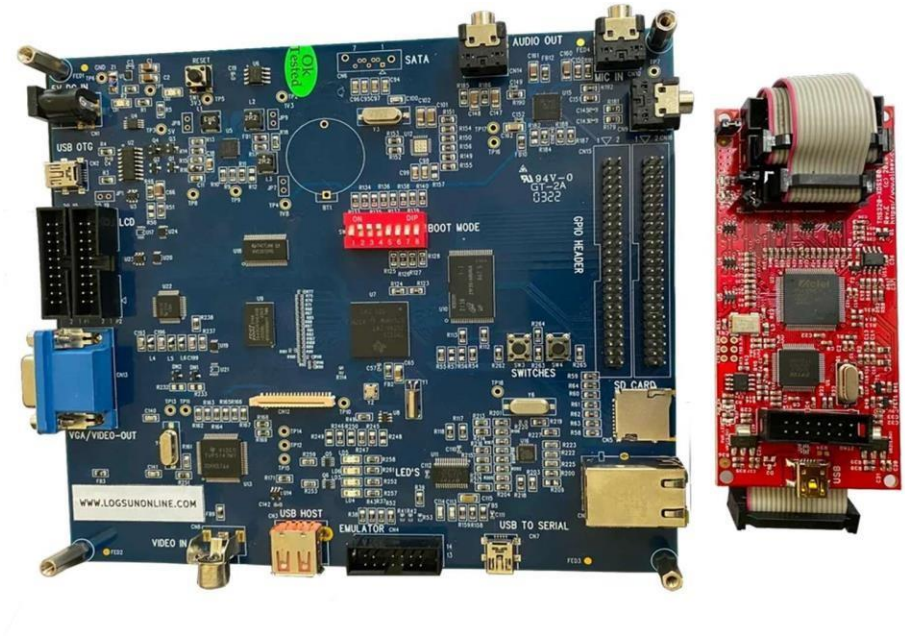
**Introduction to the Code Composer Studio**

CCS is the DSP industry's first fully integrated development environment (IDE) with DSP specific functionality. With a familiar environment like MS-based C++ TM, Code Composer lets you edit, build, debug, profile & manage projects from a single unified environment. Other unique features include graphical signal analysis, injection/extraction of data signals via I/O, multi-processor debugging, automated testing and customization via C-interpretive scripting language & much more.

**Some of the features are:**

1. IDE
2. Debug IDE
3. Advanced watch windows
4. Integrated editor
5. File I/O, Probe Points and graphical algorithm scope probes
6. Advanced graphical analysis
7. Interactive profiling
8. Automated testing & customisation via scripting

9. Visual project management system
10. Compile in background while editing & debugging
11. Multi-processor debugging
12. Help on target DSP




DSP (TMS320C6748) Development board with XDS100V3 USB Emulator

**PROCEDURE:**

1. Once the PC is booted, turn ON the power of DSP kit. Now connect the USB of DSP starter kit to the PC.
2. Start CCS
3. To create project: **File/New/CCS Project**
   - i) Select C674x Floating-point DSP
   - ii) Target - TMS320C6748
   - iii) Connection – Texas Instruments XDS100v3 USB Emulator
   - iv) Project creation name
   - v) Complier version – TIv7.4.8
   - vi) Finish
   - vii) On the top left **name [Active Debug]** will appear and **main.c** window will open.

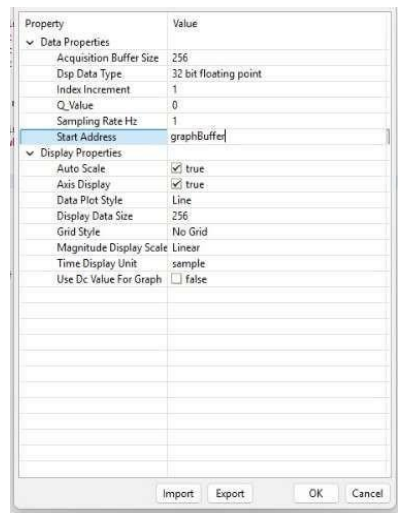
4. Now we are supposed to write a program based on ‘C’.

**Code:**
```
#include <math.h>

#define BUFFER_SIZE 256
float graphBuffer[BUFFER_SIZE];
float squareBuffer[BUFFER_SIZE];
float theta = 0;

void main(void) {
    int i;
    while(1) {
        for(i = 0; i < BUFFER_SIZE; i++) {
            graphBuffer[i] = sin(theta);
            if(graphBuffer[i] > 0)
               squareBuffer[i] = 1.0;
            else
               squareBuffer[i] = -1.0;
            theta += 0.1;
        }
        theta = 0;
    }
}
```

5. To build: **Project/Build Project**

6. To debug: **Run/Debug**

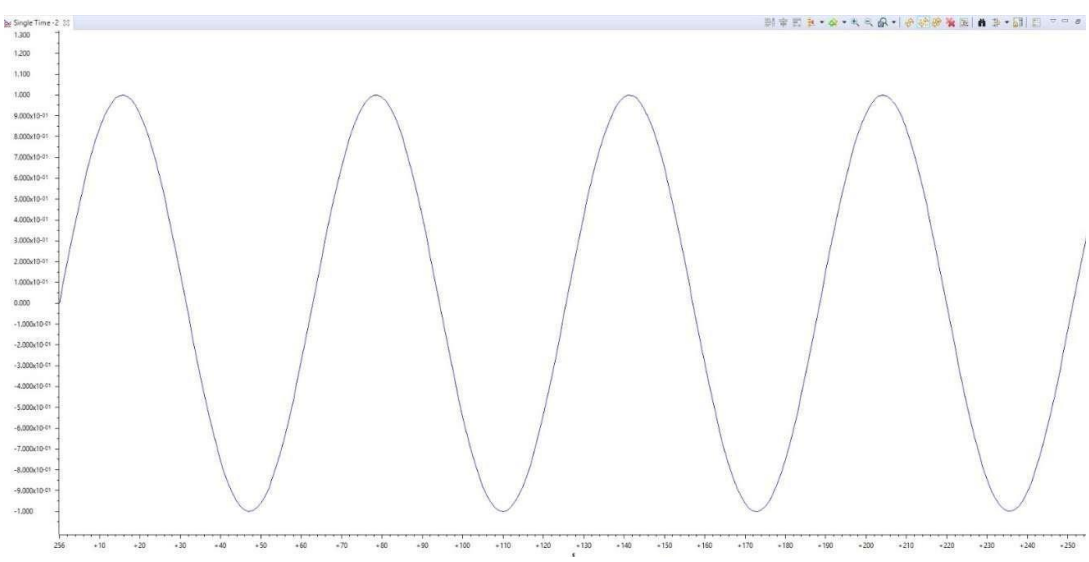
7. Tools property:



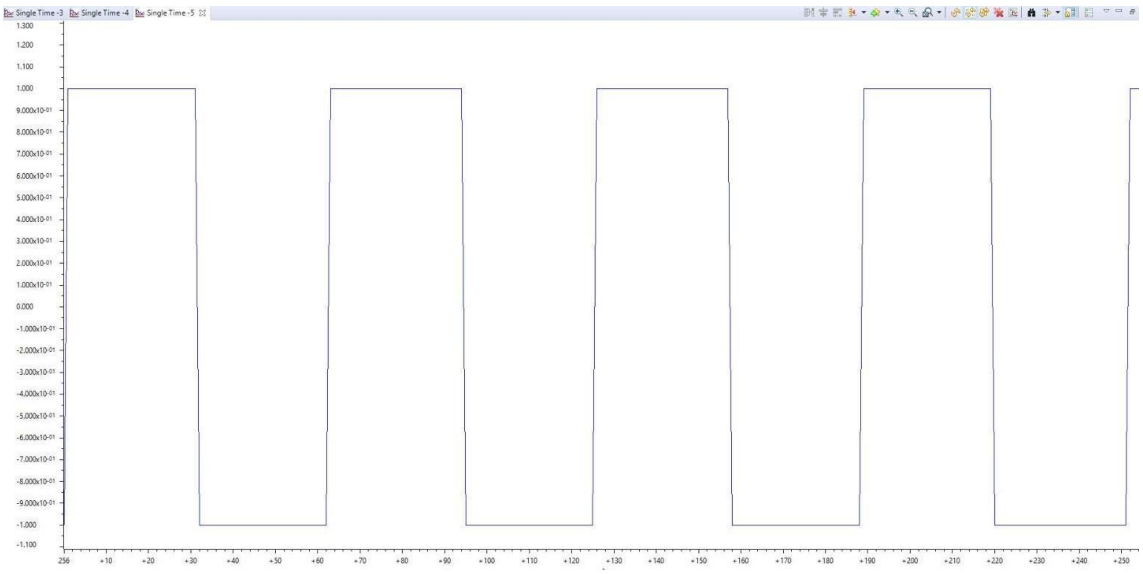

To run a project:

**Plots:**

Sine wave:



Square wave:




**Result:** We have successfully done each part of the experiment and matched the data with calculated values and understood the operation.